Softmax / Output Probabilities
Diff-Weighted CE (+ Pointer Grounding Loss)

Linear (lm_head, Tied) + Softcap (30.0)
1536 → 262.144 Vocab Logits

Final Decoder RMSNorm

DECOUPLED CAUSAL DECODER | **N_dec = 35× (E2B-IT · d=1536)**

⊕ Add & Post-FFN RMSNorm

Feed Forward (GeGLU) + PLE Gate
d_dec=1536 → d_ff=6144 → 1536
15 Cross-KV Owners (14.26M) + 20 Shared (L13/14)

⊕ o_proj (+LoRA) & RMSNorm

self_out + [tanh(cross_gate) · 2σ(gate_proj)] ⊙ cross_out

Decoupled Cross-Attn
cross_q_norm(Q) / β_N
cross_kv_proj(Z_enc)
(Stationary · Zero RoPE)
→ cross_out

Masked Self-Attention
RoPE(q_norm(Q))
RoPE(k_norm(K)), v_norm(V)
(Causal · Window 512)
→ self_out

Q_cross (No RoPE)

Q_self, K_self, V_self

Pre-RMSNorm → Shared q_proj (+ LoRA r=16)

+ Causal RoPE

Output Embedding + Per-Layer Emb (PLE)
Tied w/ LM Head · 262.144 × 1536

Outputs / Instruction (Shifted Right)
decoder_input_ids [B, N_dec ≤ 8.192]

Z_enc
K_cross, V_cross

SWIGLU DUAL-GATE BRIDGE (encoder_projection)
7,082,496 Params · Maps H_enc (1536) → Z_enc (1536)

⊕ Post-RMSNorm(d_dec) ⊙ γ_calib

SwiGLU Non-Linear
W_down(SiLU(W_g)⊙W_u)

w_res: Identity()
(0 params — Simetris)

Pre-RMSNorm(d_enc)

H_enc [B, N_enc, d_enc]

Final Encoder RMSNorm

BIDIRECTIONAL ENCODER | **N_enc = 35× (E2B · d=1536)**

⊕ Add & Post-FFN RMSNorm

Feed Forward (GeGLU) + PLE Gate
d_enc=1536 → d_ff=6144 → 1536 (Double-Wide)
+ Per-Layer Input Gate (256d)

⊕ Add & Pre/Post RMSNorm

Bidirectional GQA Self-Attention
8 Q / 1 KV · Sliding (512) & Global Full
15 KV-Owners + 20 KV-Shared
+ Encoder LoRA (r=16, α=32, FP32)

V Q K

RoPE (θ=10k/1M) +

Input Embedding + Per-Layer Emb (PLE)
262.144 × 1536 + PLE (35 × 256)

Inputs (Repository / Code Context)
encoder_input_ids [B, N_enc ≤ 32.768]